# Integrated Circuits for Wireless Communication

MINIPROJECT-3




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# Direct Conversion Receiver

Design a Direct Conversion Receiver chain for 2.45 GHz WLAN application. The chain includes LNA, quadrature mixer, filter and VGA. Filter and VGA need can be in macromodel. Ideal sources can be used to generate LO. The receiver should meet the following specifications:

- Centre frequency: 2.41 GHz to 2.49 GHz
- Channel bandwidth: 20 MHz
- Sensitivity: −85 dBm with $SNR_{\text{min}} = 6$ dB
- In-band blocker at 25 MHz offset: −65 dBm
- ADC: Input full-scale level of 0 dBm with 70 dB dynamic range

Objective is to meet the sensitivity specification in the presence of blocker while consuming minimum DC power.

## Solution

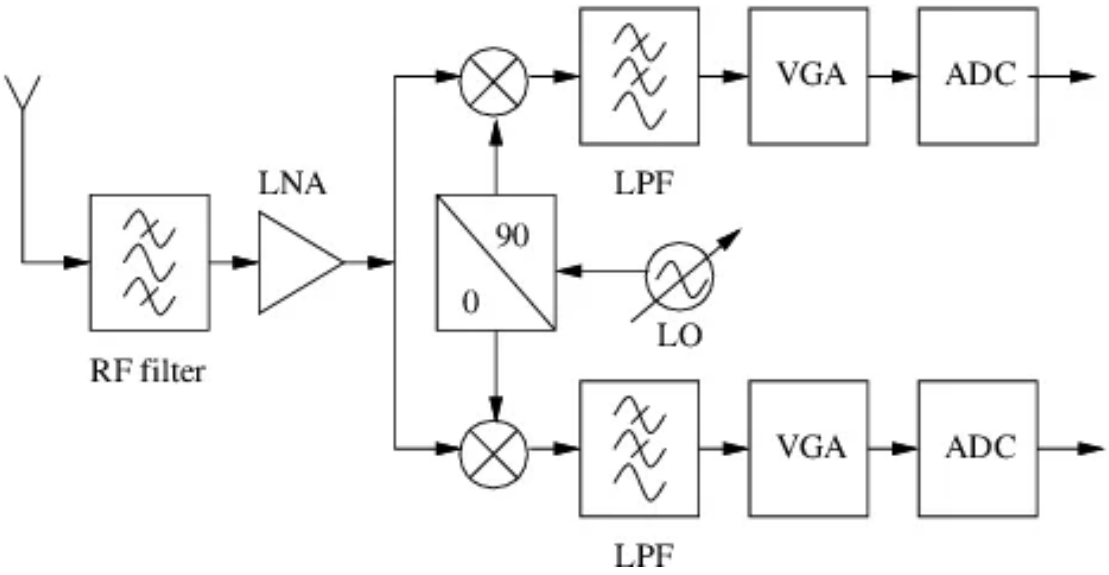



**Figure 1:** *Block Diagram of the Direct Conversion Receiver*

Using the sensitivity equation:

$$
\begin{aligned}
P_{\text{sense}} &= -174 + 10\log_{10}(B) + NF + SNR \\
-85 &= -174 + 10\log_{10}(20 \times 10^6) + NF + 6 \\
-85 &= -174 + 73 + NF + 6 \\
-85 &= -95 + NF \\
NF &= 10\,\text{dB}
\end{aligned}
$$

# 1. LNA

The target center frequency is 2.45 GHz, which gives an angular frequency of:

$$\omega_0 = 2\pi f_0 = 2\pi(2.45 \times 10^9) \approx 15.39 \times 10^9 \text{ rad/s}$$
$$\omega_0^2 \approx 237 \times 10^{18} \text{ (rad/s)}^2$$

The output tank is designed to resonate at the center frequency of 2.45 GHz. The resonant frequency is:

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Solving for $C_2$ using $L_2 = 3.5\,\text{nH}$:

$$C_2 = \frac{1}{\omega_0^2 L_2}$$
$$C_2 = \frac{1}{(237 \times 10^{18})(3.5 \times 10^{-9})}$$
$$\approx 1.20 \times 10^{-12} \text{ F} = 1.2\,\text{pF}$$

The input impedance is:

$$Z_{in} = \frac{g_m L_s}{C_t} + j\left(\omega L_g + \omega L_s - \frac{1}{\omega C_t}\right)$$

For resonance:

$$\omega L_g + \omega L_s = \frac{1}{\omega C_t}$$

Total inductance:

$$L_{in} = L_g + L_s = 9.5\,\text{nH} + 0.6\,\text{nH} = 10.1\,\text{nH}$$

Capacitance:

$$C_t = \frac{1}{\omega_0^2 L_{in}}$$
$$C_t = \frac{1}{(237 \times 10^{18})(10.1 \times 10^{-9})}$$
$$\approx 417\,\text{fF}$$

For Input Matching: Real Part (50 Ω):

$$R_{in} = \frac{g_m L_1}{C_t}$$

For $R_{in} = 50\,\Omega$:

$$50 = \frac{g_m(600 \times 10^{-12})}{417 \times 10^{-15}}$$
$$g_m = \frac{50(417 \times 10^{-15})}{600 \times 10^{-12}}$$
$$\approx 34.75\,\text{mS}$$

- 10 pF: DC blocking capacitors.
- $C_1$ (1 pF): AC bypass capacitor for the cascode gate.
- $R_0, R_1, R_3, R_4$: High-value biasing resistors (100 kΩ to 1 MΩ).

## 2. Mixer

The mixer schematic is based on double-balanced Gilbert cell topology. To maximize the transit frequency ($f_T$) and minimize parasitic capacitances at 2.45 GHz, the channel length for all core transistors is set to:

$$L = 45\,\text{nm}$$

The gate bias for the RF stage is:

$$V_{G,RF} = 0.6\,\text{V}$$

The overdrive voltage is:

$$\begin{aligned} V_{OV,RF} &= V_{GS,RF} - V_{th} \\ &= 0.6 - 0.35 \\ &= 0.25\,\text{V} \end{aligned}$$

$$g_{m,RF} = \mu_n C_{ox} \left(\frac{W}{L}\right)_{RF} V_{OV,RF}$$

Assuming $W_{RF} = 60\,\mu\text{m}$:

$$\left(\frac{W}{L}\right)_{RF} \approx \frac{60\,\mu\text{m}}{45\,\text{nm}} \approx 1333$$

This ensures a large transconductance in the range of:

$$g_{m,RF} \approx 14\text{–}20\,\text{mS}$$

Also, LO bias is:$V_{G,LO} = 1.0\,\text{V}$
The LO transistor width is chosen as:

$$W_{LO} = \frac{1}{2} W_{RF} = 30\,\mu\text{m}$$

The voltage conversion gain is:

$$A_v = \frac{2}{\pi} g_{m,RF} R_D \qquad ; \quad R_D = 1\,\text{k}\Omega$$

$$\begin{aligned} A_v &= 20 \log_{10} \left(\frac{2}{\pi} \cdot 0.0142 \cdot 1000\right) \\ &\approx 19.2 dB \end{aligned}$$

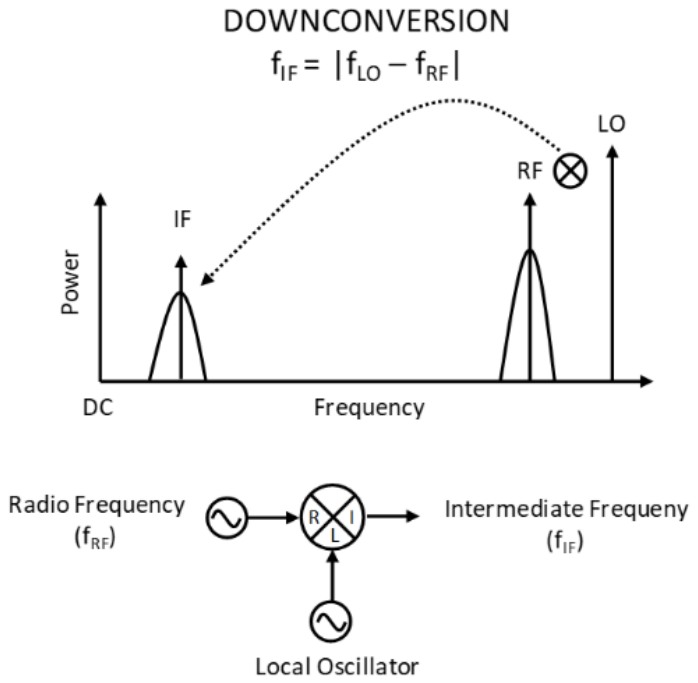



**Figure 2:** *Schematic of Low Noise Amplifier*

After the mixer, the output is passed through Low Pass Butterworth Filter with appropriate corner frequency is used to extract our desired baseband signal and remove unwanted harmonics from the spectrum and passed to the Variable Gain Amplifier which is implemented in the schematic using an ideal VCVS. The VGA adjusts the signal amplitude to match the 0dBm full scale input requirement of the ADC.

## Schematic:

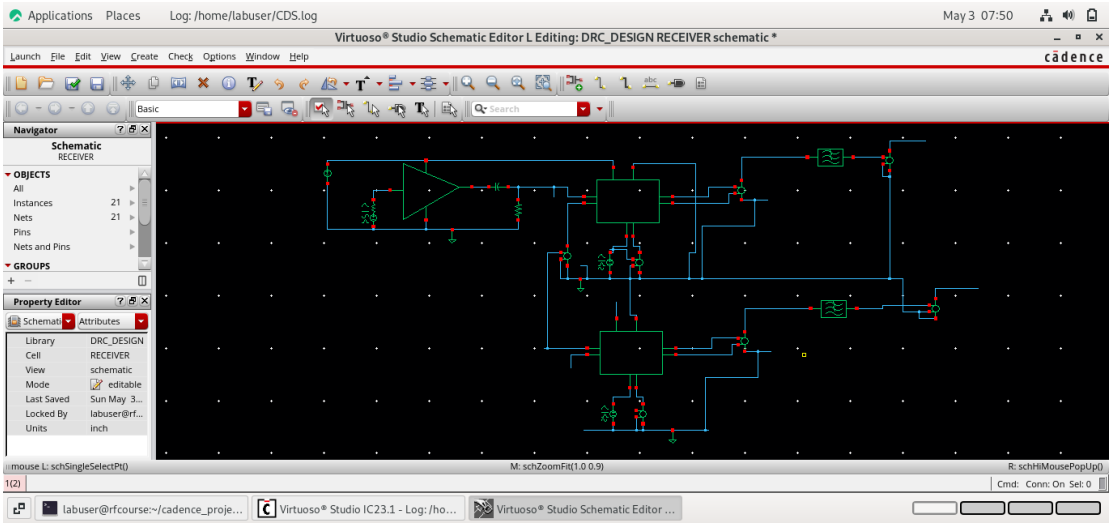



**Figure 3:** *Schematic of Direct Conversion Receiver*

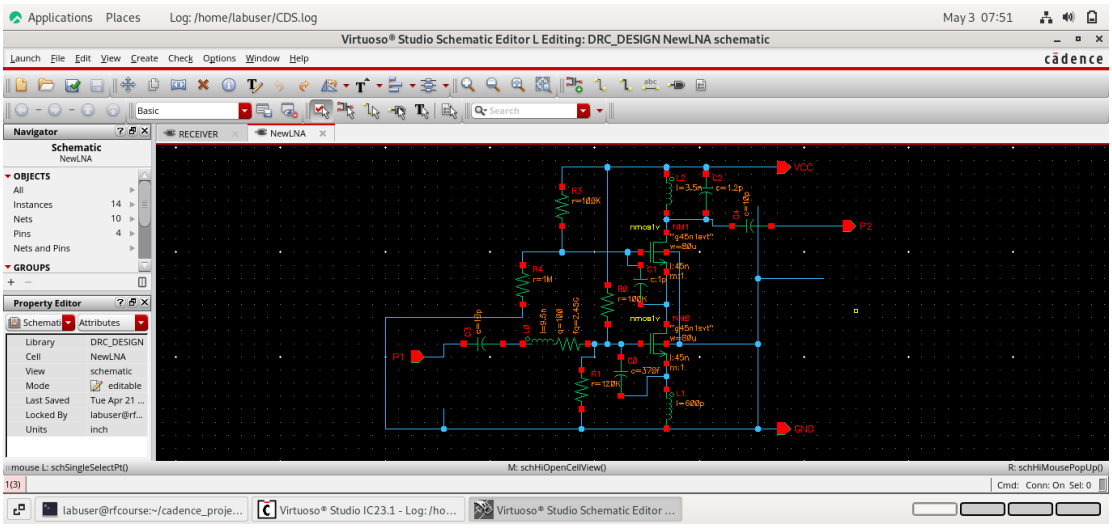



**Figure 4:** *Schematic of Low Noise Amplifier*

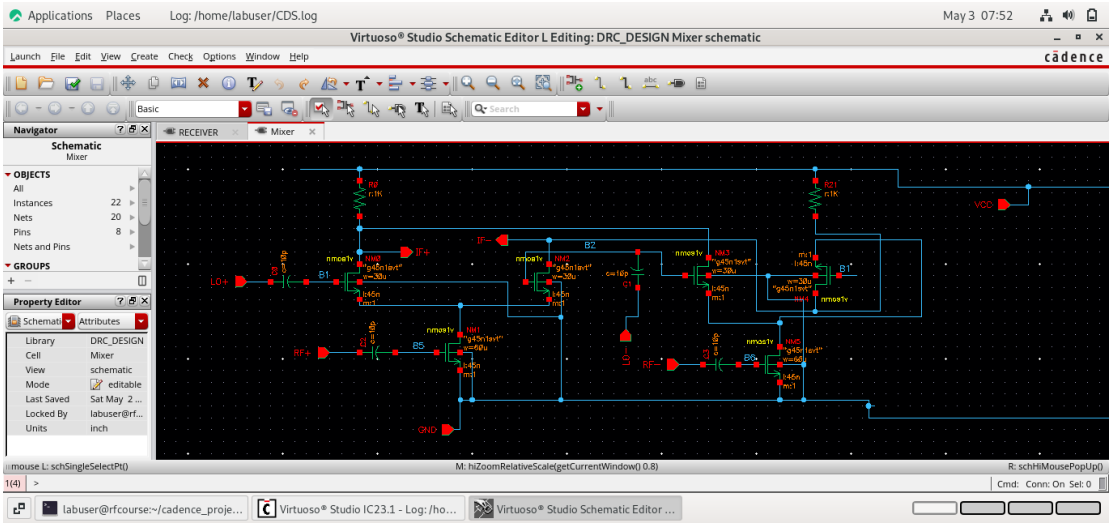
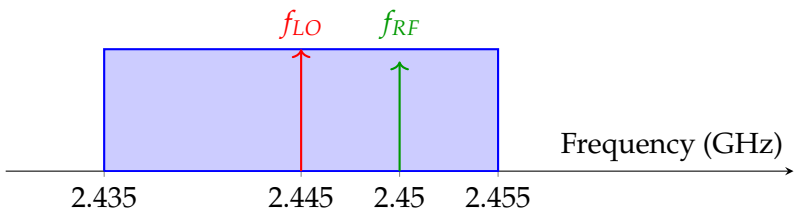



**Figure 5:** *Schematic of Double Balanced Active Gilbert Mixer*

## 0.1 Simulation Results and Analysis (Single Tone)

Initial simulations were done using a single-tone input to get the fundamental gain, noise, and compression results of the receiver chain.



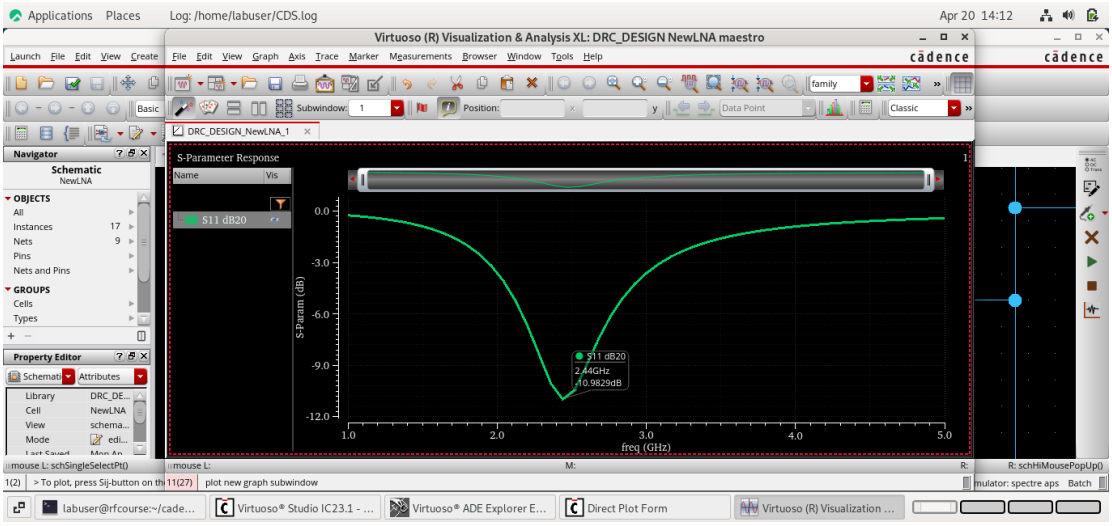



**Figure 6:** *S11 showing the matching of LNA*

### 0.1.1 Periodic AC (PAC) Response

As shown in Fig. 7 and Fig. 8, a PAC analysis was performed to determine the conversion gain of the receiver chain.

- The receiver achieves a voltage conversion gain of 25.52 dB at the 5 MHz IF. Since, we used VGA of gain 2 we can subtract 6dB out of it to get the approximate conversion gain of mixer(19.2dB).
- The response at the 2.45 GHz RF center frequency shows the down-conversion to the 5 MHz baseband signal.
- The Inphase and Quadrature phase signals achieved similar conversion gains.

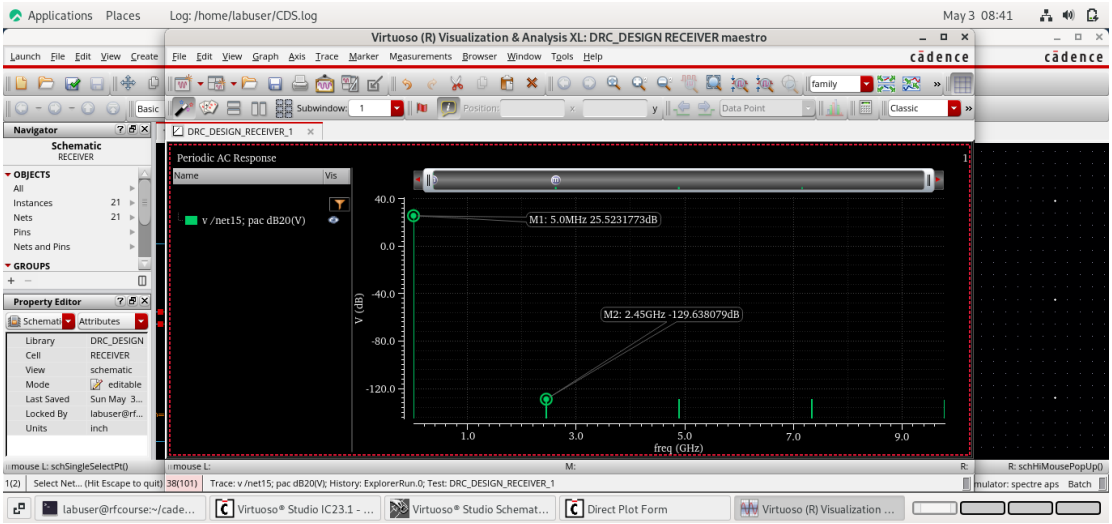



**Figure 7:** *Periodic AC Response of the Inphase signal*

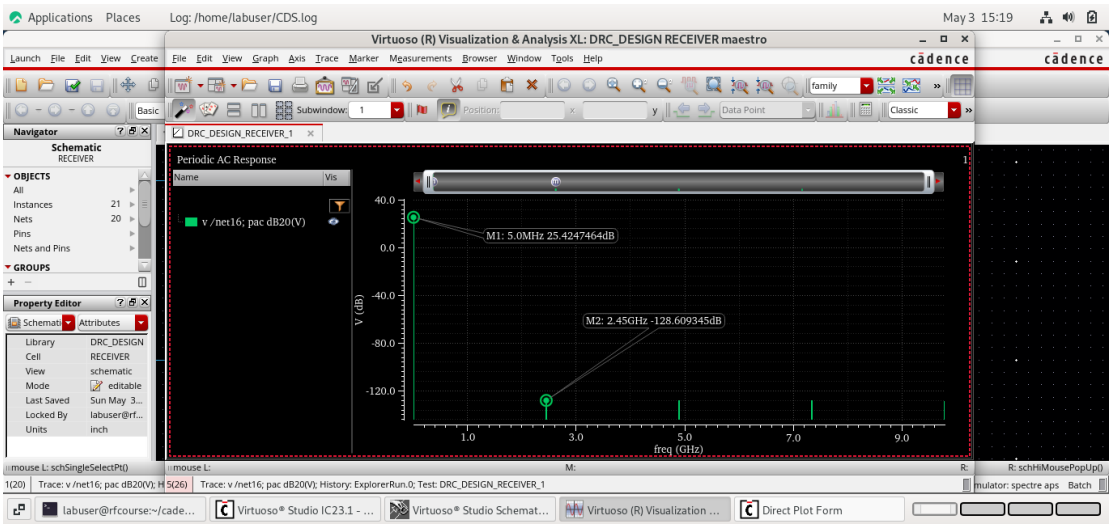



**Figure 8:** *Periodic AC Response of the Quadrature phase signal*

### 0.1.2 Noise Performance (Pnoise)

Sensitivity is directly limited by the Noise Figure (NF) of the receiver chain.

**Sensitivity Calculation:**

$$\begin{aligned} P_{\text{sense}} &= -174 + 10\log_{10}(20 \times 10^6) + NF + SNR \\ &= -174 + 73 + 5.7 + 6 \\ &= \mathbf{-89.3}\,\text{dBm} \end{aligned}$$

This exceeds the required $-85$ dBm sensitivity by 4.3 dB.

- From the Periodic Noise (pnoise) simulations (Fig. 9 and Fig. 10) similar Noise Figure of approximately 5.62 dB & 5.62681 dB in Inphase and Quadrature phase networks.

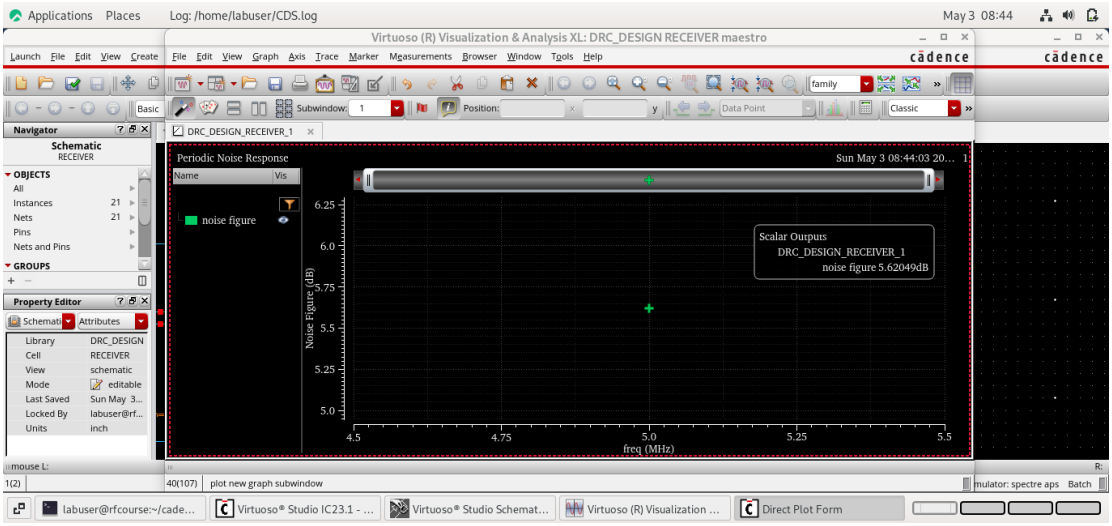



**Figure 9:** *Noise Figure of the Inphase signal at $f_{IF}$ frequency*

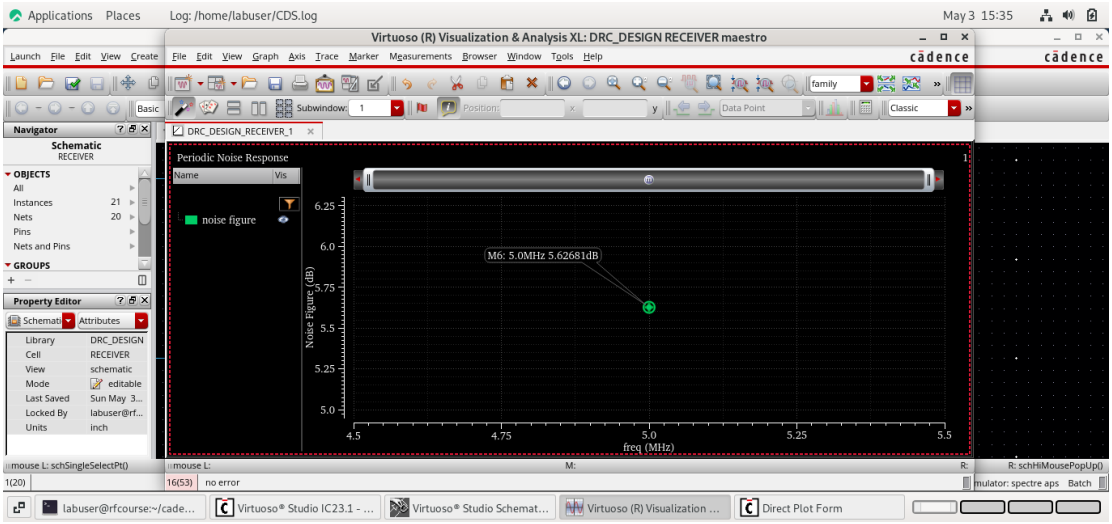



**Figure 10:** *Noise Figure of the Quadrature phase signal at $f_{IF}$ frequency*

### 0.1.3 Compression Point ($P_{1dB}$)

To determine the upper limit of the signal power before saturation, the 1 dB compression point was simulated.

Using the Harmonic Balance Analysis,

- Input Referred $P_{1\text{dB}}$: As seen in Fig. 11 and Fig. 12, the input $P_{1\text{dB}}$ is approximately:

$$\text{Full Receiver} \approx -40.18\ \text{dBm}$$

- Compression across both In-phase (I) and Quadrature (Q) paths are similar for balanced performance.

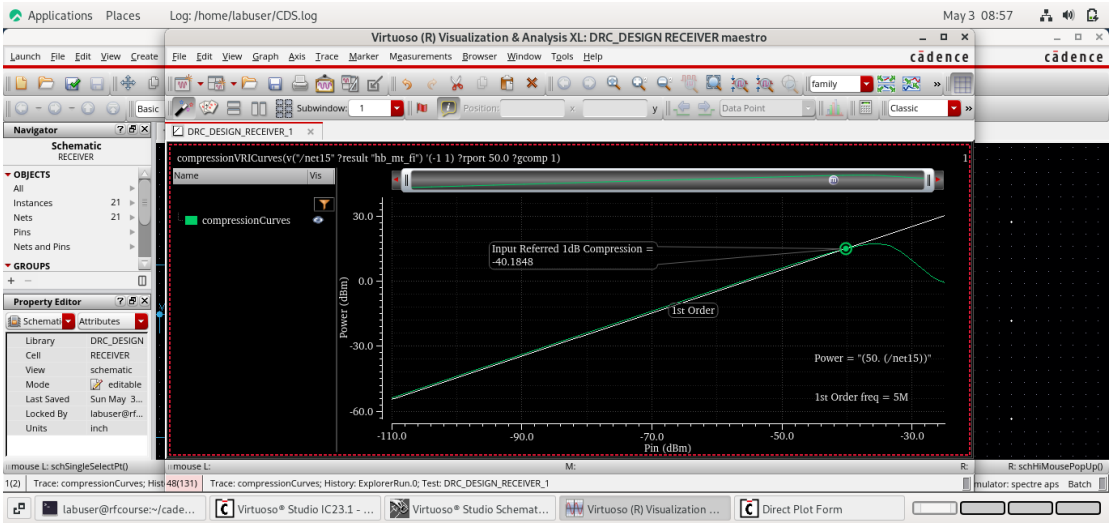



**Figure 11:** *Compression point of the Inphase signal*

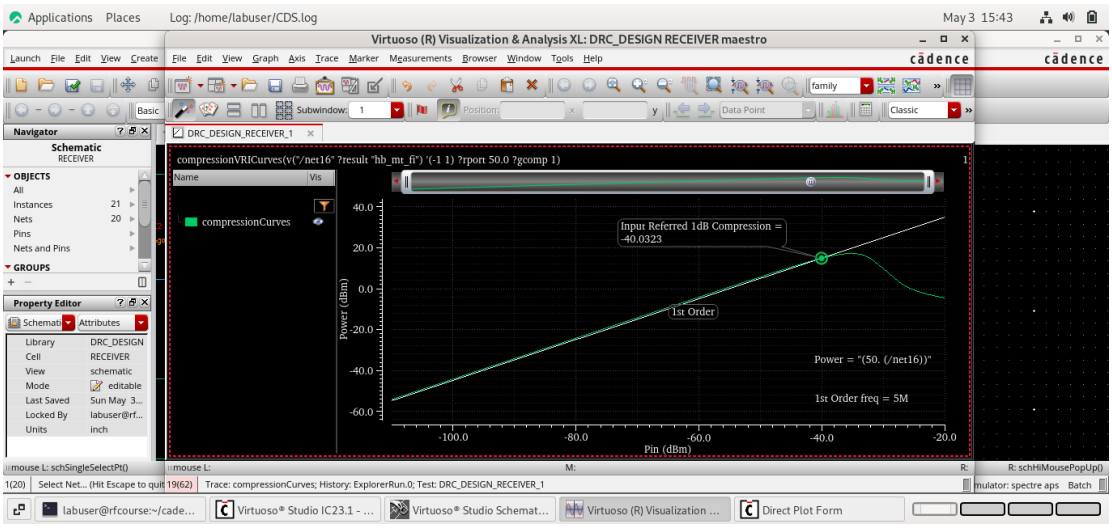
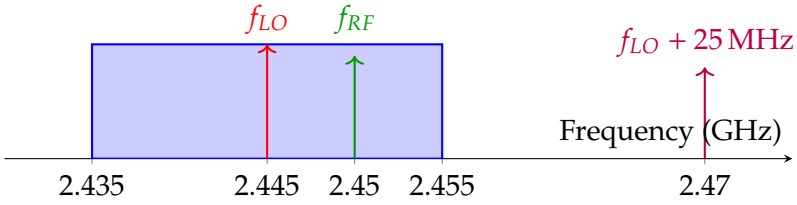



**Figure 12:** *Compression point of the quadrature phase signal*

## 0.2 Simulation and Analysis (Double Tone)




For the Intermodulation Distortion (IMD) simulation of the mixer, the two input RF frequencies are chosen such that the resultant third-order intermodulation (IM) products, after down-conversion, lie within the IF bandwidth. The two input frequencies are:

$$\begin{aligned} f_1 &= 2.45\ \text{GHz} \quad ;\ \text{desired RF signal} \\ f_2 &= 2.47\ \text{GHz} \quad ;\ \text{blocker signal} \end{aligned}$$

$$\begin{aligned} f_{\text{LSB}} &= 2f_1 - f_2 \\ &= (2 \times 2.450) - 2.470 \\ &= 2.430\ \text{GHz} \end{aligned}$$

$$\begin{aligned} f_{\text{USB}} &= 2f_2 - f_1 \\ &= (2 \times 2.470) - 2.450 \\ &= 2.490\ \text{GHz} \end{aligned}$$

The Local Oscillator frequency is:

$$f_{LO} = 2.445\ \text{GHz}$$

After frequency down-conversion, the corresponding IF frequencies of the third-order IM products become:

$$\begin{aligned} |f_{\text{LSB}} - f_{LO}| &= |2.430 - 2.445| \\ &= 15\ \text{MHz} \end{aligned}$$

and

$$\begin{aligned} |f_{\text{USB}} - f_{LO}| &= |2.490 - 2.445| \\ &= 45\ \text{MHz} \end{aligned}$$

To ensure proper operation in the presence of strong interference, the receiver was tested against out-of-channel blockers.

### 0.2.1 Blocker Rejection

The specification requires handling a −65 dBm blocker at a 25 MHz offset.

- The PAC response in Fig. 13 shows approximately **100 dB** rejection at 25 MHz offset.

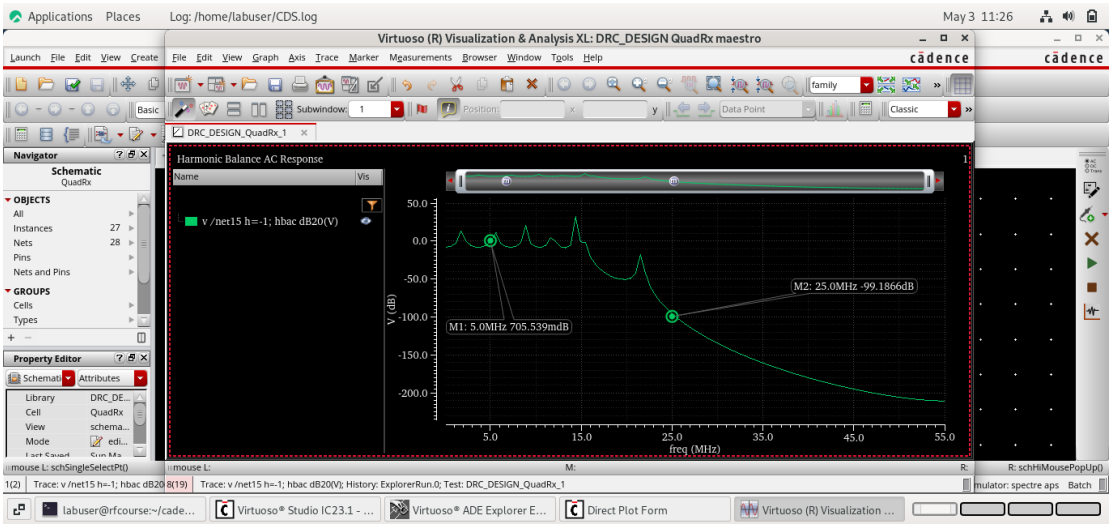



**Figure 13:** *HBAC Analysis plot showing the rejection of blocked tone*

The lower sideband IM product appears at 15 MHz, which is exactly at the edge of the designed 15 MHz Butterworth filter bandwidth.

The HBAC sweep shows that the 15 MHz Butterworth filter provides approximately 100 dB of rejection at the 25 MHz blocker offset. Since, the blocker is attenuated by 100 dB, it reaches the output significantly weaker than the −85 dBm desired signal, preventing desensitization.

### 0.2.2 Third-Order Intercept Point (IIP3)

A two-tone Harmonic Balance simulation was used to evaluate linearity.

- **Input Referred IP3:** As shown in the simulation results,

$$\text{IIP3} \approx -32.75\,\text{dBm}$$

- **IM3 Product Estimation:**

$$P_{\text{IM3}} \approx -129.5\,\text{dBm}$$

This is significantly below the sensitivity floor, ensuring minimal distortion.

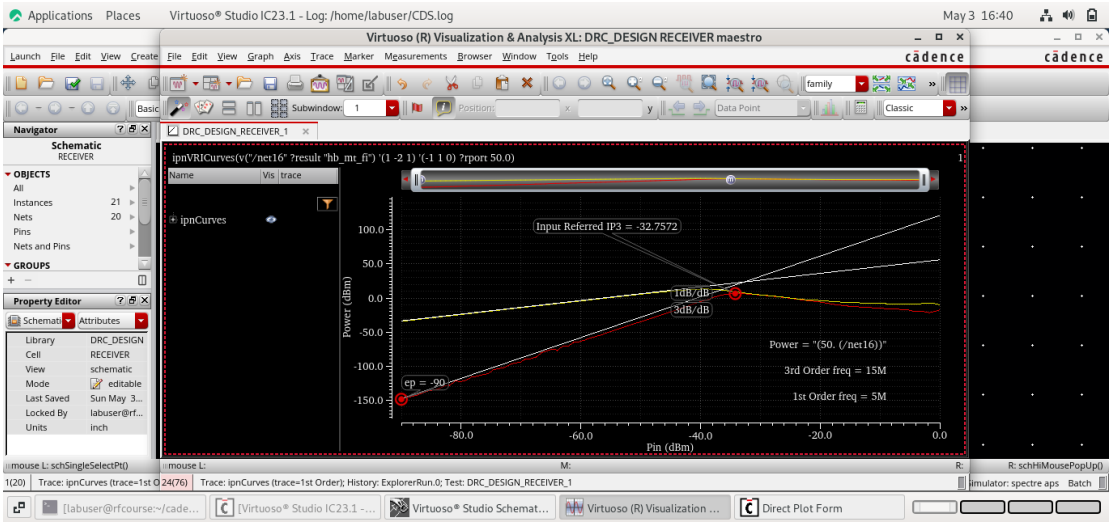



**Figure 14:** *IIP3 plot at 3rd order freq = 15MHz (Lower side)*

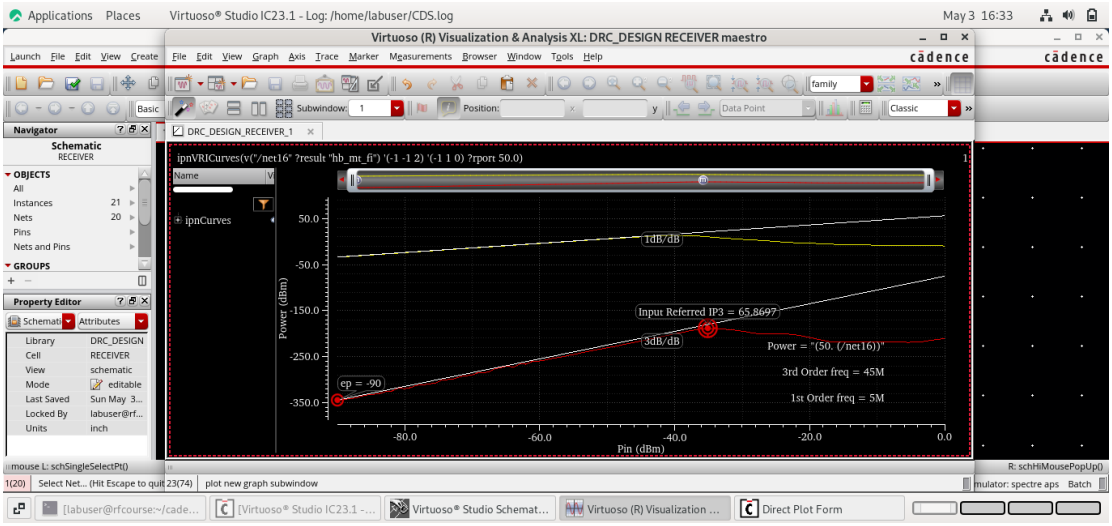



**Figure 15:** *IIP3 plot at 3rd order freq = 45MHz (Upper side)*

## 0.3 Observations:

- **$IM_3$ Calculation:**

$$\begin{aligned} P_{IM3,in} &= 3 \times P_{blocker} - 2 \times IIP3 \\ &= 3(-65) - 2(-32.75) \\ &= -195 + 65.5 \\ &= -129.5\,\text{dBm} \end{aligned}$$

  Since $-129.5\,\text{dBm} \ll -91\,\text{dBm}$, the distortion is negligible and will not desensitize the receiver.
- The downconverted signal must fit within the ADC limits (0 dBm full scale, 70 dB dynamic range).
  Signal Levels at ADC Input:

$$\begin{aligned} \text{Desired Signal} &= -85 + 19.2 = -65.8\,\text{dBm} \\ \text{Blocker (before filtering)} &= -65 + 19.2 = -45.8\,\text{dBm} \end{aligned}$$

  With Filter Rejection (100 dB):

$$\begin{aligned} P_{blocker,ADC} &= -65 + 19.2 - 100 \\ &= -145.8\,\text{dBm} \end{aligned}$$

  The blocker is effectively eliminated, and the desired signal lies well within the ADC dynamic range without approaching saturation.

## Conclusion

From the above observations and plots we can conclude that:

- The receiver achieves a sensitivity of −89.3 dBm, which is 4.3 dB better than the required −85 dBm specification.
- Through the use of a 15 MHz Butterworth filter, the design provides approximately 100 dB of rejection at the 25 MHz blocker offset.
- The measured IIP3 of −32.75 dBm is sufficient to keep third-order intermodulation products ($P_{IM3}$) at −129.5 dBm, far below the noise floor of −91 dBm.
- The conversion gain and filtering has set the desired signal reach the ADC at approximately −65.8 dBm, within the 70 dB dynamic range and far below the 0 dBm full-scale limit.